

DATA-DRIVEN EDUCATION MONITOR

Project Documentation · Academic Year 2025–2026

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Subject Correlations

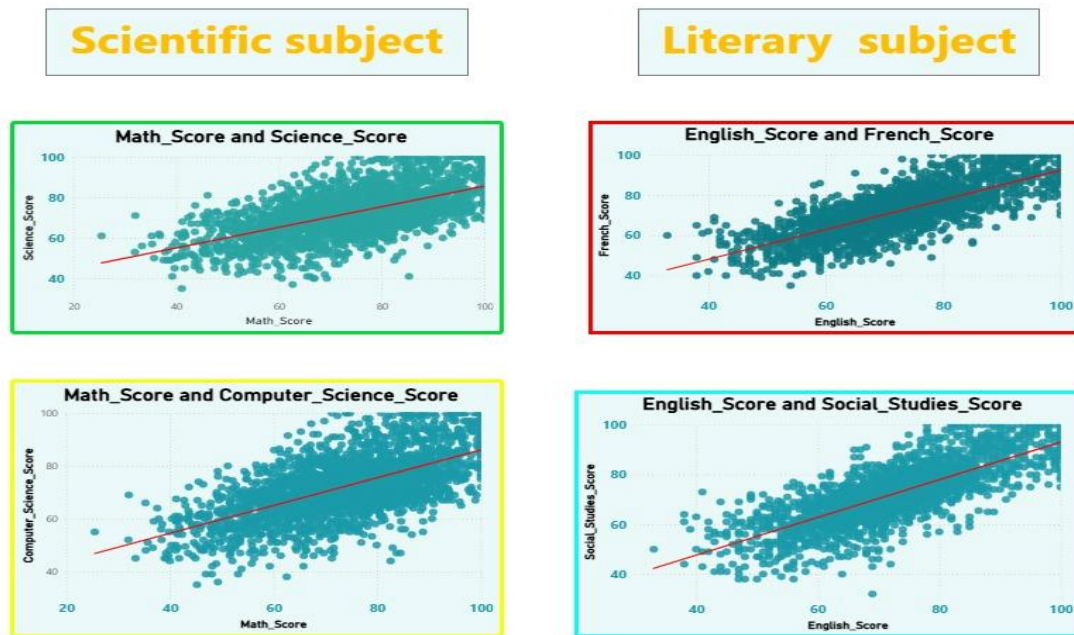


Figure 1 — Data-Driven Education Monitor: Navigation Overview of All 9 Dashboard Pages

1. Executive Summary

The Data-Driven Education Monitor is a comprehensive, end-to-end data analytics platform designed to transform raw student data into actionable intelligence for educators and administrators. Built on the Microsoft Fabric ecosystem and powered by Apache Spark, the system ingests, cleans, and models records for 1,356 students across three school departments, surfacing insights across nine interconnected Power BI dashboard pages.

Beyond traditional reporting, the project integrates an AI-powered chatbot combining Retrieval-Augmented Generation (RAG) with a Pandas analytical agent, enabling natural language interrogation of the educational dataset. Together, the BI dashboards and chatbot provide a 360-degree view of student performance — encompassing academic scores, socio-economic background, wellbeing, digital engagement, and behavioral compliance.

Project Scope & Scale

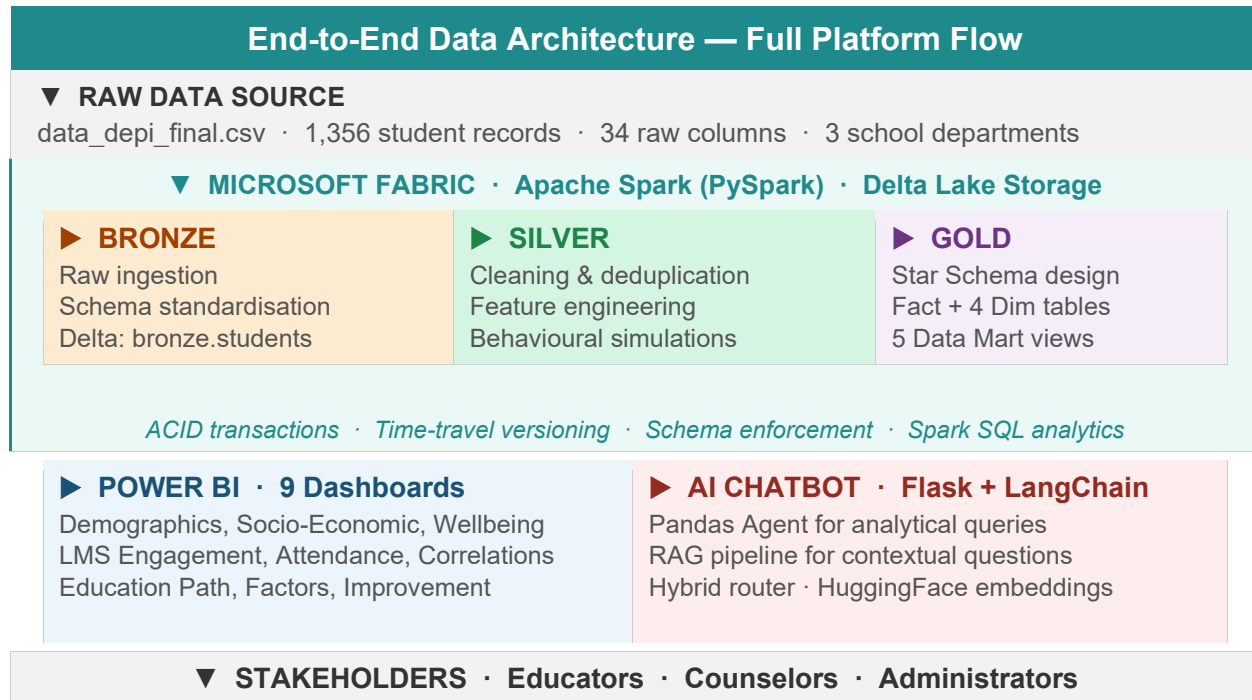
Dataset: 1,356 student records · 34 raw features · 3 school departments Pipeline: 3-tier Medallion Architecture (Bronze → Silver → Gold) · 5 analytical Data Mart views Dashboards: 9 Power BI pages · 35+ distinct visualizations AI Chatbot: Hybrid RAG + Pandas Agent · Flask backend · HTML/JS frontend Project positioning: An industry-style prototype demonstrating enterprise data engineering, business intelligence, and applied AI on a unified modern data platform.

2. Technology Stack

Component	Technology
Cloud Platform	Microsoft Fabric
Processing Engine	Apache Spark (PySpark)
Storage Format	Delta Lake · ACID transactions · Time-travel versioning
Query Language	Spark SQL + Python
BI & Visualization	Microsoft Power BI
NLP Framework	LangChain
Embedding Model	Hugging Face sentence-transformers
Vector Store	FAISS / ChromaDB
Backend API	Flask (Python)
Frontend	HTML + CSS + JavaScript

3. System Architecture Overview

The diagram below illustrates the complete end-to-end data flow of the platform, from raw CSV ingestion through the three Medallion layers on Microsoft Fabric, into Power BI dashboards and the AI chatbot, and finally to end-user stakeholders.



4. Data Pipeline: Medallion Architecture on Microsoft Fabric

The ETL pipeline follows the Medallion Architecture, organizing data into three progressive quality tiers. All tables are persisted as Delta Lake tables to guarantee ACID transactions, time-travel, and schema enforcement.

4.1 Technical Metrics & Pipeline Specifications

Metric	Detail	Value
Dataset size	Student records ingested from CSV	1,356 rows
Feature count	Raw columns before standardisation	34 columns
Schema renames	Columns renamed in Bronze layer	34 mappings
Delta tables	Total Delta tables created	Bronze(1) + Silver(1) + Gold(6)
Data Mart views	Specialised analytical Delta views	5 views
Engineered features	Simulated behavioral attributes	6 features
Star Schema tables	Fact + Dimension tables in Gold	1 Fact + 4 Dim
BI Dashboard pages	Power BI report pages	9 pages
Total visualizations	Distinct charts and AI visuals	35+

4.2 Bronze Layer — Raw Data Ingestion

The Bronze layer performs faithful ingestion with minimal transformation, establishing a permanent audit trail of source data.

- Source: CSV file (data_depi_final.csv) loaded into a Spark DataFrame via PySpark.
- Schema Standardization: A mapping dictionary renamed all 34 raw columns into clean, consistent names (e.g., First_Name, Student_ID, Math_Score, Family_Income_Level).
- Storage: DataFrame written as Delta Table (bronze.students) enabling full ACID compliance and row-level versioning from the first layer.

4.3 Silver Layer — Data Cleaning & Behavioral Simulation

A. Data Quality Operations

- Deduplication: dropDuplicates() removed any redundant records before enrichment.

- Null Audit: Comprehensive null-value analysis performed across all 34 features to ensure integrity prior to feature engineering.
- Formatting: All numeric metrics rounded to two decimal places for reporting consistency.

B. Advanced Feature Engineering (6 Simulated Attributes)

Six behavioral and socio-economic attributes were simulated using domain-driven logical constraints, not random values:

- Internet Access: Derived from Family_Income_Level via probability logic — High income (~95%), Medium (~75%), Low (~35%).
- Academic Attendance: Modeled from Total_Avg with tiered attendance bands (e.g., Avg $\geq 85 \rightarrow 90\text{--}100\%$ attendance).
- Study Hours: Assigned from Grade category (e.g., Grade A $\rightarrow 26\text{--}35$ hrs/week, Grade F $\rightarrow 1\text{--}5$ hrs/week).
- LMS Engagement: lms_login_freq_per_day and resource_access_avg_hrs scaled by academic performance and internet availability.
- Sleep Hours: Modeled per grade to reflect academic pressure patterns observed in educational research.
- Stress Level: Calculated by custom formula: $\text{Stress} = 10 - ((\text{Sleep} - 4) / 5) \times 8 + \text{Noise}$. Late assignments directly linked to resulting stress score.

4.4 Gold Layer — Star Schema & Analytical Data Marts

The Gold layer organizes cleaned data into a Star Schema optimized for Power BI DirectQuery performance.

Table / View	Description
Fact_Performance	All quantitative measures: subject scores, attendance, stress, LMS hours, late assignments
Dim_Identity	Student demographics: Name, Age, Gender
Dim_Academic_Context	Academic status: Department, Grade, Extracurricular Activities
Dim_Social	Socio-economic background: Family Income, Parent Education Level
Dim_Assignments	Assignment submission metrics: on-time vs. late counts by student
Mart_Demographics	Analytical view: performance breakdown across gender and departments
Mart_Socioeconomic	Analytical view: family income vs. digital engagement correlation
Mart_Study_Stress	Analytical view: study hours, sleep, and stress relationship by grade

Mart_LMS	Analytical view: LMS platform activity vs. final academic scores
Mart_Attendance	Analytical view: attendance compliance vs. late submission patterns

5. Design Justification: Why We Chose These Technologies

This section documents the reasoning behind key architectural decisions. Every technology was chosen deliberately based on engineering requirements, not convention. These justifications are also answers to the questions most commonly asked in project defenses.

5.1 Data Engineering Decisions

Q: Why did you choose the Medallion Architecture?

A: The Medallion Architecture separates concerns across three layers with clear, enforceable quality contracts. The Bronze layer preserves a raw, immutable copy of source data — essential for auditability and reprocessing if business logic changes. The Silver layer enables complex transformations and behavioral simulations without corrupting raw data. The Gold layer organizes clean data into a Star Schema optimized for analytical query patterns. This progressive quality model is the standard approach in modern data lakehouses and directly mirrors how enterprise platforms like Databricks and Microsoft Fabric are structured in industry.

Q: Why did you choose Delta Lake as the storage format?

A: Delta Lake was chosen over plain Parquet or CSV for three critical capabilities: (1) ACID transactions guarantee no partial writes corrupt the table during pipeline runs; (2) time-travel versioning allows querying any historical snapshot of the data — essential for debugging and reproducibility; and (3) schema enforcement prevents upstream schema changes from silently corrupting downstream tables. For a multi-layer pipeline where each layer reads from the previous one, these guarantees are architectural necessities, not optional features.

Q: Why use Apache Spark (PySpark) when the dataset is only 1,356 rows?

A: Three reasons: First, scalability — the pipeline is designed to scale. The same PySpark code that processes 1,356 rows processes 1.3 million rows without code changes, only infrastructure changes. Building in Pandas would require a complete rewrite at scale. Second, platform alignment — Microsoft Fabric is natively a Spark-based platform; PySpark is the idiomatic and supported approach, not an overengineering choice. Third, SQL expressiveness — Spark SQL enables the analytical views and Star Schema decomposition to be expressed as declarative, maintainable queries rather than imperative row-by-row Python code.

5.2 Business Intelligence Decisions

Q: What is the most important KPI in the project, and why?

A: The most strategically important KPI is Academic Attendance Rate by Grade. It is the single variable most strongly correlated with academic outcome across the entire dataset —

A-grade students maintain 95% attendance while F-grade students average only 39%. Unlike study hours or LMS logins (which are private behaviors), attendance is directly observable and actionable by administrators in real time. It serves as both a leading indicator (early attendance decline predicts grade decline) and a lagging indicator (poor grades correlate with retrospective attendance gaps).

Q: How would an actual school use this dashboard system?

A: Three concrete use cases: (1) Early Warning System — administrators monitor Attendance & LMS Engagement dashboards weekly; any student whose login frequency drops below 1 per day or attendance falls below 75% triggers a counselor check-in, intervening before grades deteriorate. (2) Resource Allocation — the Socio-Economic dashboard identifies students lacking home internet, enabling targeted offline resource distribution or lab access scheduling. (3) Pathway Guidance — counselors use the Education Path radar chart to identify each student's strongest subject cluster and advise on STEM vs. Humanities tracks, backed by data rather than subjective impression.

5.3 AI Chatbot Decisions

Q: Why did you use RAG instead of fine-tuning an LLM?

A: Fine-tuning would require substantial compute resources, a curated training dataset, and produces a model frozen at training time — unable to incorporate new data without retraining. RAG retrieves relevant documents at inference time from a vector store that can be updated incrementally. This means the chatbot's knowledge base can be extended with new cohort data, updated policies, or new research documents without any model retraining. For an educational system where data changes each semester, RAG is the architecturally correct and cost-efficient choice.

Q: What is the difference between the Pandas Agent and the RAG module? How does routing work?

A: The Pandas Agent handles structured, analytical queries — questions requiring computation against the student DataFrame (averages, counts, comparisons, filtering). It translates natural language into executable Pandas operations so the LLM can query structured data without manually written code. The RAG module handles unstructured, contextual queries — explanatory or descriptive questions where the answer comes from educational documents, not the dataset. The Hybrid Router distinguishes the two pathways by detecting analytical keywords (average, total, count) combined with DataFrame column names (Math_Score, Grade, Family_Income_Level) in the user query. Matches go to Pandas Agent; all others go to RAG. This dual-pathway architecture ensures every query is handled by the most appropriate engine.

6. Power BI Dashboards

The Power BI layer consists of nine interconnected dashboard pages, each serving a distinct analytical purpose and connected to the Gold layer Data Mart views. Below is a detailed walkthrough of each page with its key insights.

6.1 Academics & Demographics

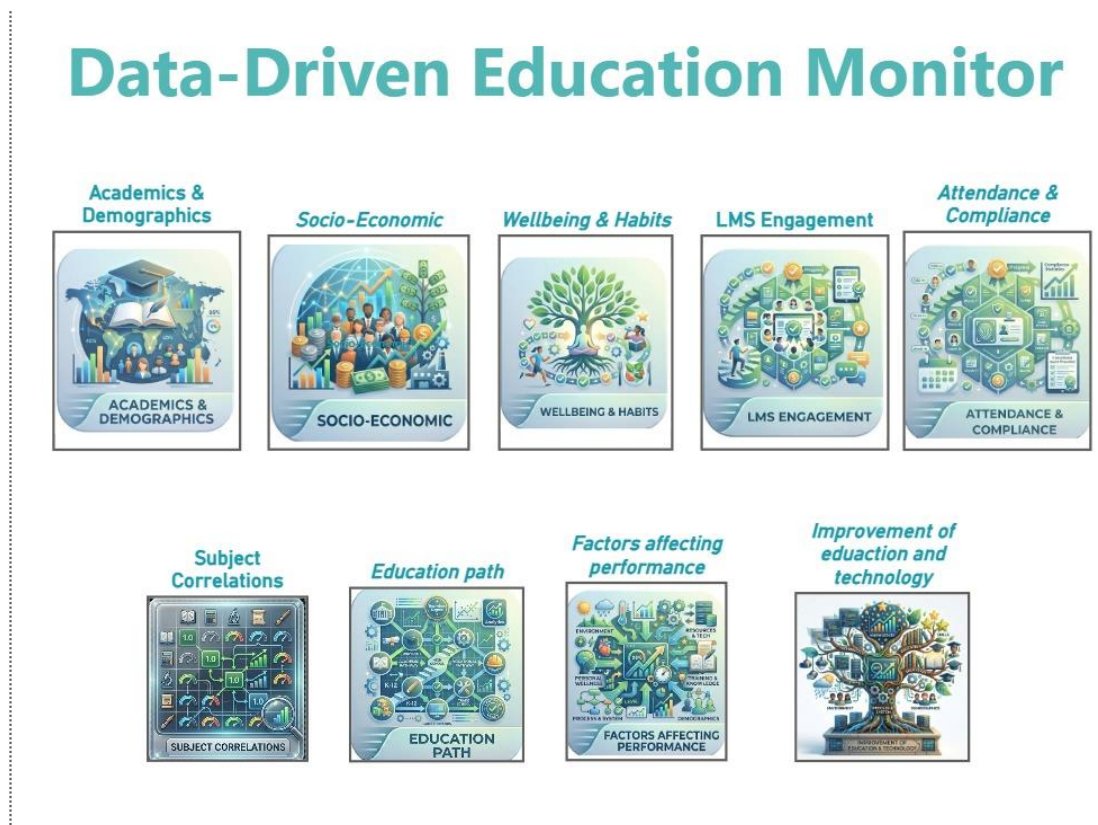


Figure 2 — Academics & Demographics Dashboard

- Total population: 1,356 students · Gender: 50.45% Male / 49.55% Female — unbiased gender sampling.
- Department distribution: 6th Grade (828), 5th Grade (822), Middle School (798) — near-uniform, enabling fair cross-department comparisons.
- Peak academic performance: Middle School Math average = 85, highest single subject-department combination in the entire dataset.
- Extracurricular baseline: ~57% of students across all departments do not participate in extracurricular activities — explored further in the Wellbeing dashboard.

6.2 Socio-Economic Overview

Academics & Demographics

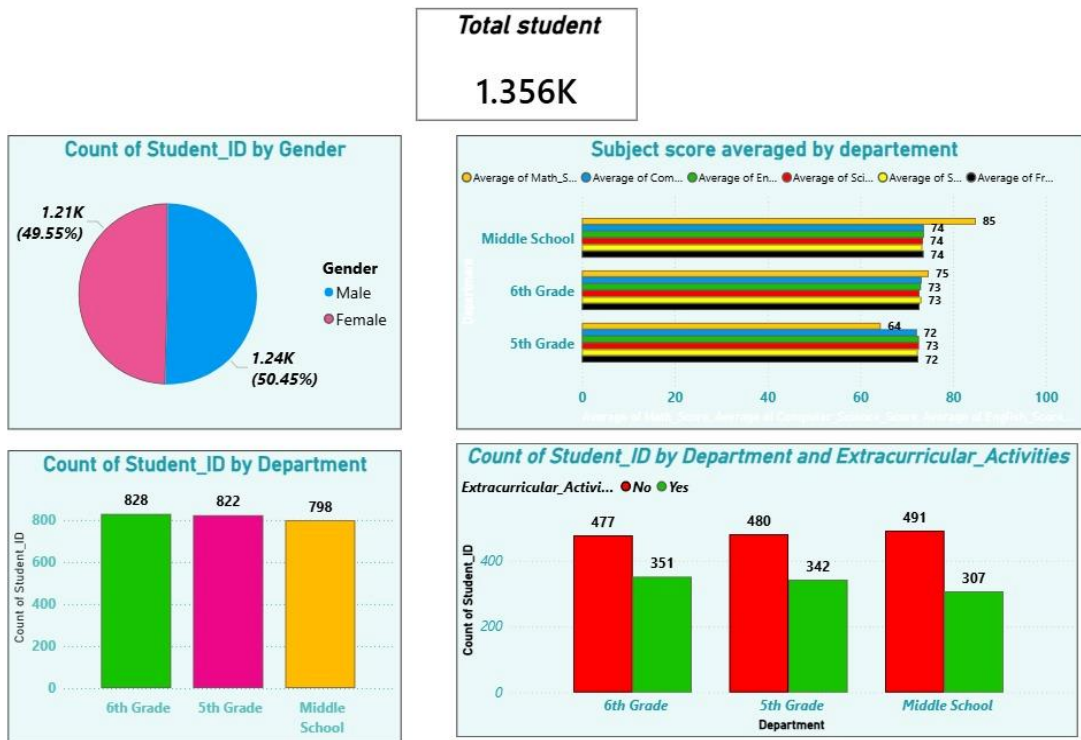


Figure 3 — Socio-Economic Overview Dashboard

- Digital divide: High-income students have 93.2% home internet access vs. only 34.34% for low-income students — a 2.72× disparity driven purely by income.
- LMS access gap: Students with home internet average 1.86 active LMS hours/day vs. 0.62 hrs/day without — a 3× engagement gap independent of academic motivation.
- Parent education is evenly distributed (PhD: 628, Bachelor's: 620, High School: 602, Master's: 598), enabling cross-dimensional analysis without selection bias.

6.3 Wellbeing & Habits

Socio-Economic overview

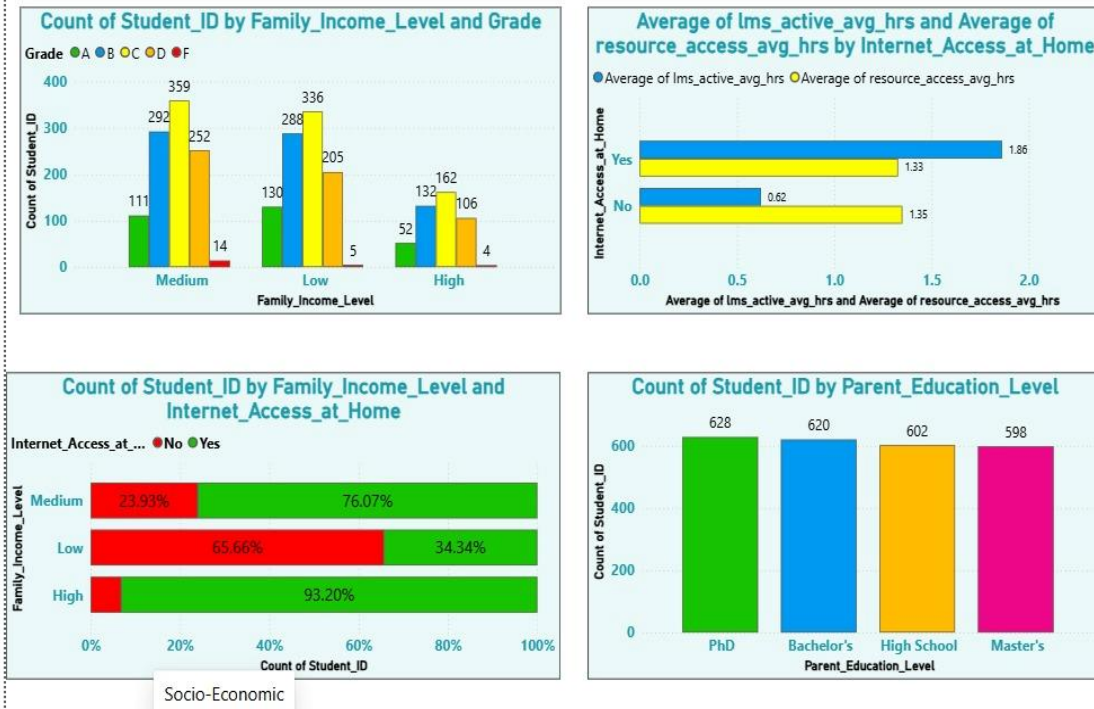


Figure 4 — Wellbeing & Habits Dashboard

- Sleep-stress inverse: Near-perfect linear relationship from stress = 1–2 at 9 hrs sleep/night to stress = 9–10 at 4–5 hrs/night.
- Stress by grade: A avg = 3.18 · F avg = 9.13 — a 2.87× difference confirming academic failure and extreme stress form a self-reinforcing cycle.
- Study hours gap: Grade A = 31 hrs/week vs. Grade F = 4 hrs/week — a 7.75× difference.
- Extracurricular paradox: Participants study more (+5 hrs/week) and report lower stress (–1.16 pts), disproving the common distraction hypothesis.

6.4 LMS Engagement

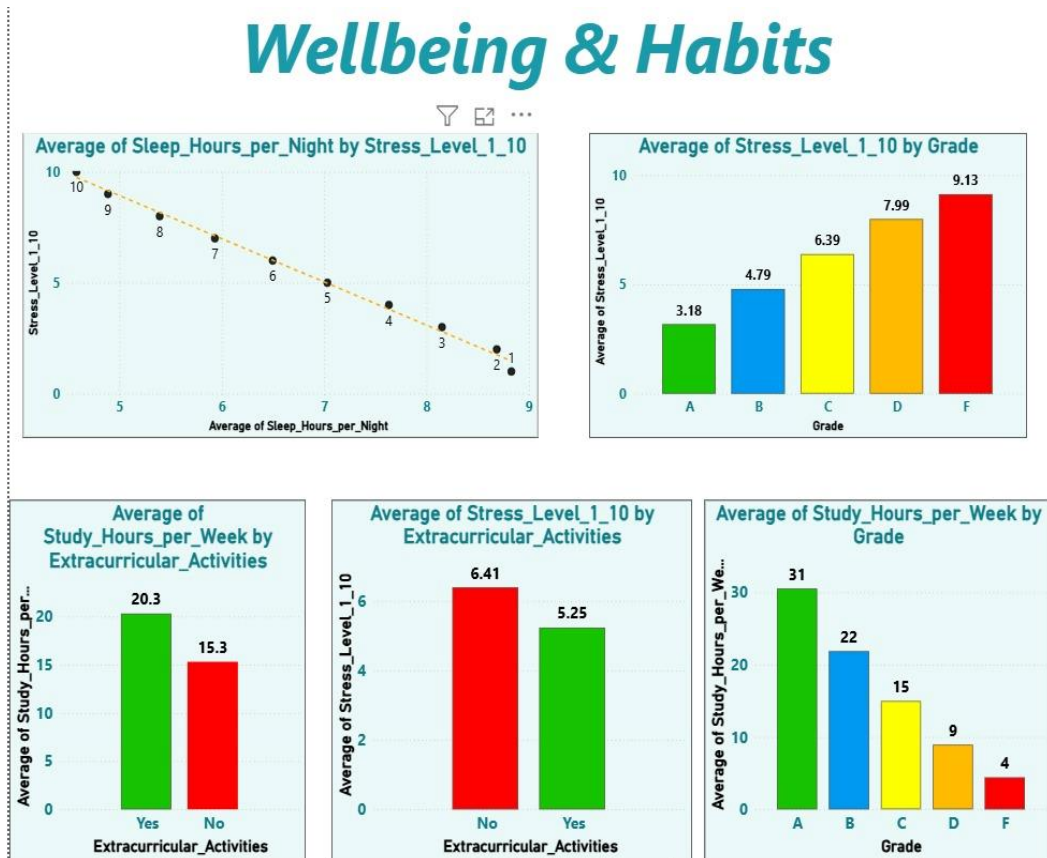


Figure 5 — LMS Engagement Dashboard

- Login frequency by grade: A = 2.8/day, B = 2.2, C = 1.5, D = 0.9, F = 0.4 — a 7× gap between top and bottom performers.
- Resource access paradox: Grade B students consume the most LMS resources (1,208 total hrs) vs. Grade A's 703 hrs, indicating A-students achieve more with greater efficiency.
- All six subject averages move in parallel with LMS active hours, confirming digital engagement is a universal academic multiplier across all disciplines.

6.5 Attendance & Compliance

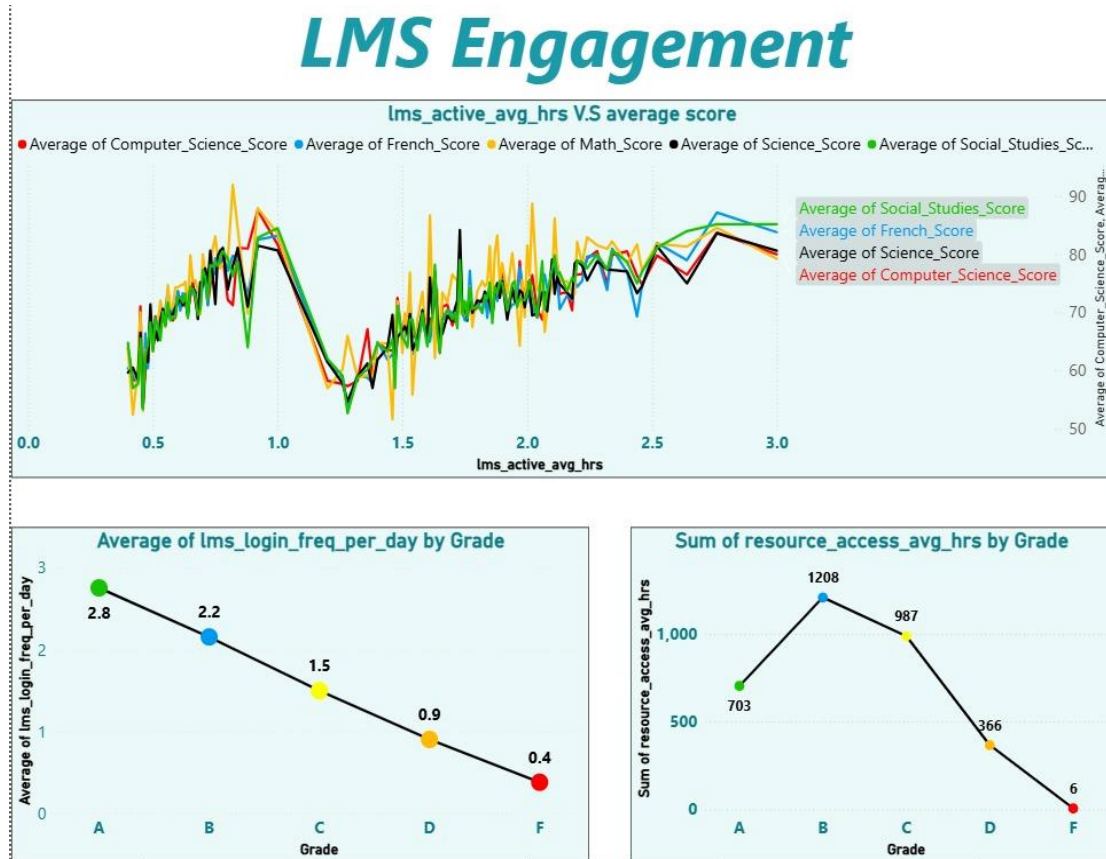


Figure 6 — Attendance & Compliance Dashboard

- Attendance by grade: A = 95%, B = 88%, C = 78%, D = 62%, F = 39% — the strongest single predictor of academic outcome in the dataset.
- LMS-compliance link: 0 late assignments correlates with 2+ daily logins; 5 late assignments correlates with fewer than 1 login/day.
- Stress-compliance link: Students at stress levels 8–10 submit 3–4 late assignments on average, confirming stress is both a cause and consequence of non-compliance.

6.6 Subject Correlations

Attendance & Compliance

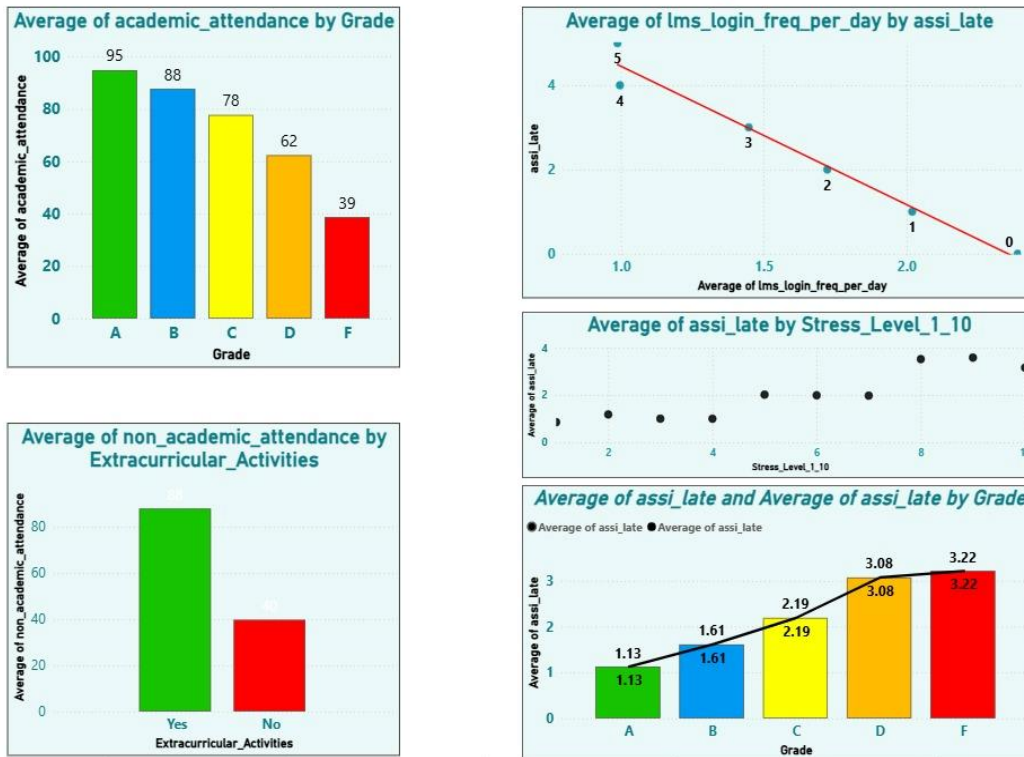


Figure 7 — Subject Correlations: Assignment Scores vs. Final Grades

Subject Correlations

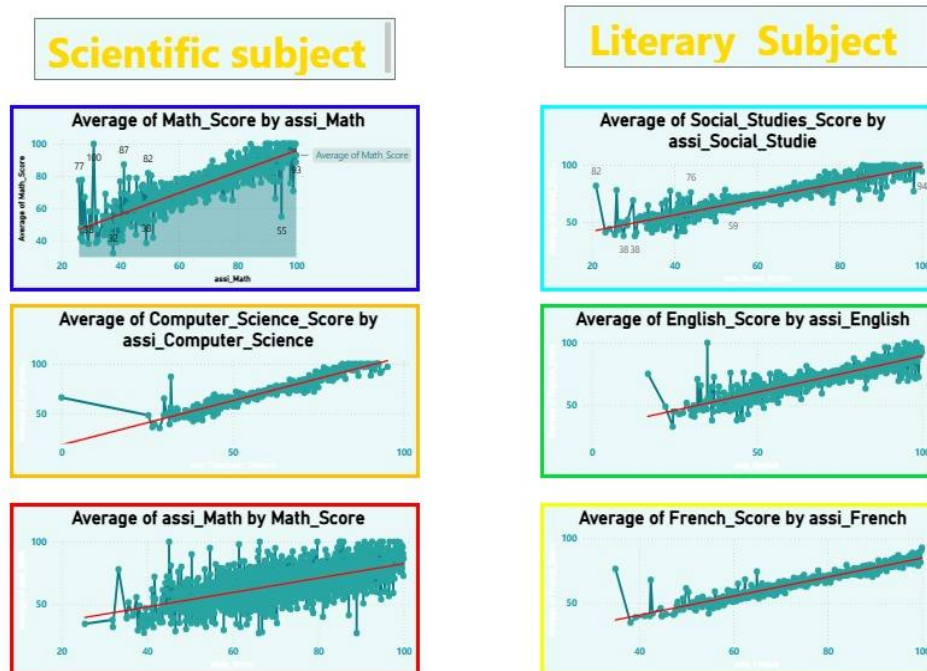


Figure 8 — Subject Correlations: Scientific and Literary Aptitude Clusters

- Assignment-to-final-grade: Strong positive correlation across all six subjects — confirming that continuous assessment is a reliable predictor of final outcomes.
- Scientific cluster: Math score reliably predicts Science and Computer Science — a shared quantitative aptitude cluster.
- Literary cluster: English score reliably predicts French and Social Studies — a shared language-processing aptitude cluster.
- Operational implication: Early assignment performance can identify at-risk students and trigger interventions before final exams.

6.7 Education Path

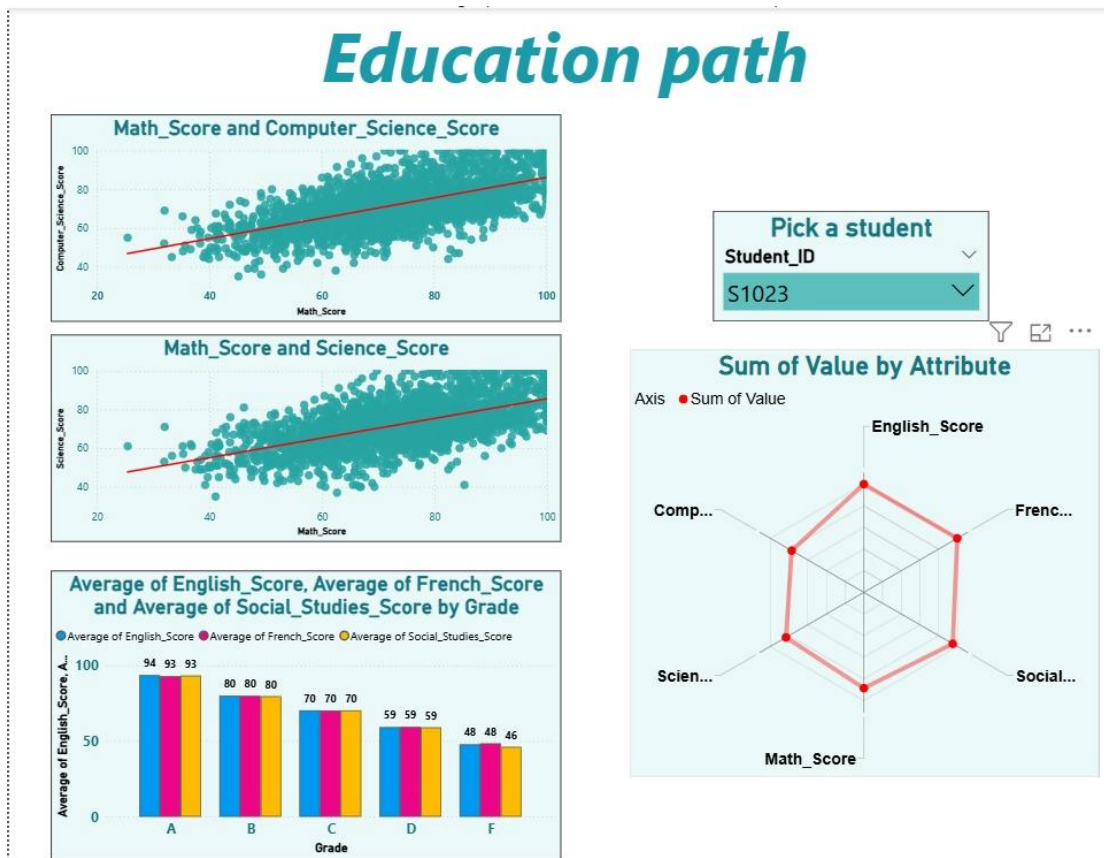


Figure 9 — Education Path: Individual Student Profiling

- Grade benchmarks: Grade A averages 94 in all three literary subjects; Grade F averages 46–48 — a consistent linear decline across the full grade spectrum.
- Radar chart: Built using Power Query data unpivoting (subject columns → attribute-value pairs), presenting a hexagonal view of each student's relative strengths across all six subjects.
- Pick-a-Student slicer: Counselors select any Student_ID to instantly generate a quantitative academic profile, enabling data-backed STEM vs. Humanities pathway guidance.

6.8 Factors Affecting Performance

Factors affecting performance

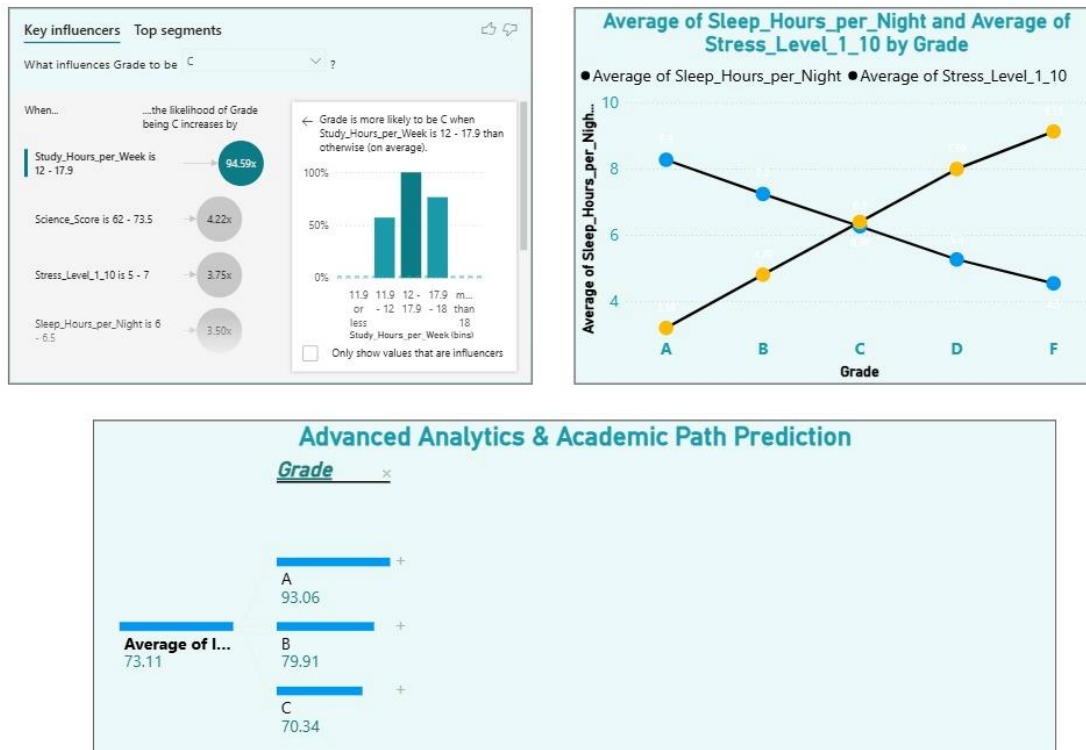


Figure 10 — Factors Affecting Performance: AI-Powered Analytics

- **Key Influencers AI visual:** Identifies that studying 12–17.9 hrs/week is a 94.59× multiplier for achieving Grade C — a precise, actionable threshold for targeted academic support.
- **Sleep-stress tipping point:** Grade C represents the physiological boundary (avg 6.5 hrs sleep, stress = 6.4) where lifestyle deterioration begins to materially affect academic outcomes.
- **Decomposition Tree:** Enables ad-hoc drill-down root-cause analysis on any aggregate metric across multiple dimensions simultaneously.
- **Predictive grade averages:** A = 93.06, B = 79.91, C = 70.34 — a quantitative classification framework for student risk stratification and early intervention.

6.9 Improvement of Education & Technology

Improvement of education and technology

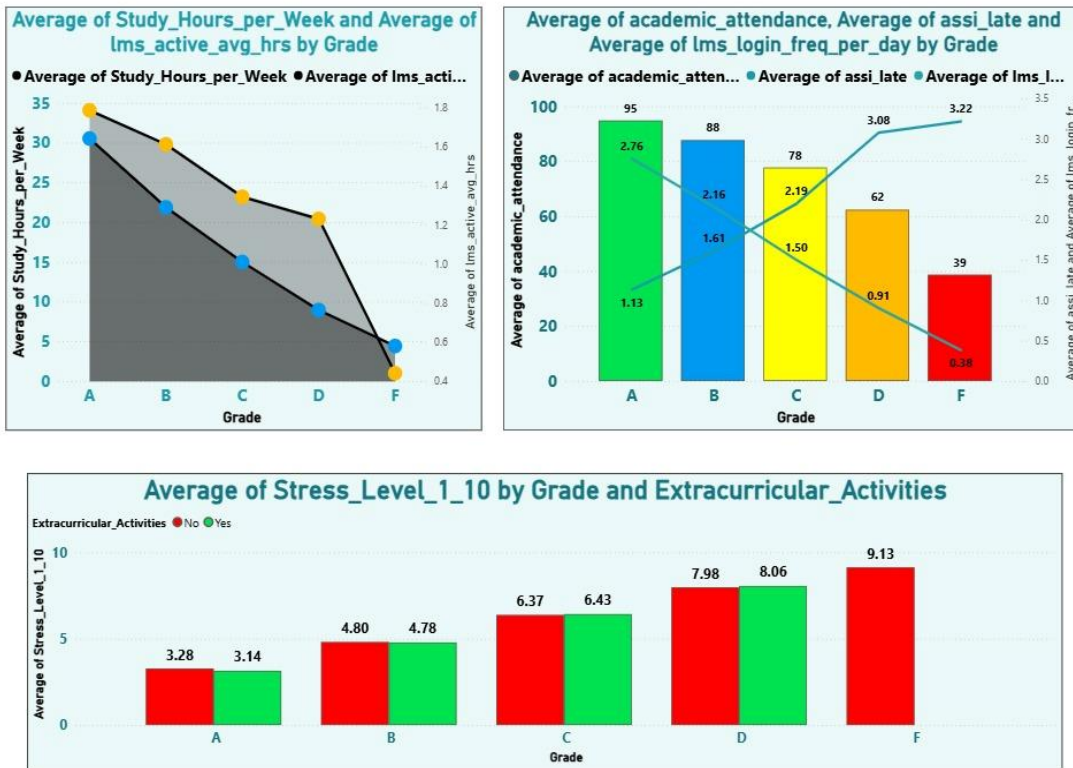


Figure 11 — Improvement of Education and Technology Dashboard

- Compound collapse: Study hours (31 → 4, a 7.75× drop) and LMS active hours (1.7 → 0.4, a 4.25× drop) decline together from Grade A to F — physical and digital disengagement are inseparable.
- Triple decline: Attendance (95% → 39%), LMS login (2.76 → 0.38/day), and late submissions (1.13 → 3.22) all deteriorate in lockstep, proving disengagement is holistic.
- Extracurricular buffer: Across every grade band, students with extracurricular activities show equal or marginally lower stress than non-participants — confirming structured engagement acts as a psychological buffer.
- Strategic conclusion: Technology, attendance, and mental health must be addressed as a unified intervention, not as separate silos.

7. AI-Powered Educational Chatbot

The AI chatbot enables any stakeholder, regardless of BI expertise, to interrogate the educational dataset using natural language. It combines a Pandas analytical agent and a RAG document retrieval module into a unified hybrid system served via a Flask web application.

7.1 System Architecture

Component	Role
Data Layer	Silver Delta table loaded into Pandas DataFrame for real-time analytical processing
LangChain	Orchestrates all LLM interactions, prompt management, and both agent pipelines
Pandas Agent	Translates natural language into Pandas operations; handles all numerical/analytical queries
RAG Module	Semantic retrieval from vector store; handles all contextual and descriptive queries
Hybrid Router	Analyzes query intent; routes to Pandas Agent or RAG based on keyword + column detection
Flask Backend	API endpoint receives user messages, invokes pipeline, returns responses in real time
HTML/JS Frontend	Asynchronous chat interface; displays responses without page reloads

7.2 RAG Pipeline

- **Text Chunking:** Educational documents segmented into smaller chunks for optimal retrieval granularity and contextual accuracy.
- **Embedding:** Hugging Face sentence-transformer models convert chunks into high-dimensional vector representations.
- **Vector Store:** Embeddings persisted in FAISS/ChromaDB enabling sub-second semantic similarity search at inference time.
- **Grounded Response:** User query is embedded, compared to stored vectors, top-ranked chunks retrieved, and supplied to the LLM as context — reducing hallucination and improving factual accuracy.

7.3 Hybrid Routing Logic

Query Routing Decision

PANDAS AGENT route — Query contains: analytical keywords (average, count, total, percentage, compare, how many) AND at least one DataFrame column name (Math_Score, Grade, Family_Income_Level, lms_login_freq_per_day, etc.) RAG route — All other queries: descriptive, explanatory, or conceptual questions not referencing structured dataset columns. This dual-pathway design ensures every query — whether asking for a specific number or a contextual explanation — is handled by the most appropriate engine, maximising both analytical precision and contextual accuracy.

8. Key Analytical Findings

The six findings below represent the most strategically significant insights across all nine dashboard pages, combining quantitative precision with actionable business implications.

#	Finding
1	Low-income students are 2.72× less likely to have home internet access, suppressing their LMS engagement by up to 3× — a structural barrier independent of academic motivation or effort.
2	Grade A students study 7.75× more hours per week (31 vs. 4) and maintain 2.44× higher academic attendance (95% vs. 39%) than Grade F students.
3	Extracurricular participation is positively correlated with both higher study hours (+5 hrs/week) and lower stress levels (−1.16 pts), directly disproving the widely-held distraction hypothesis.
4	The Power BI Key Influencers AI visual identified that studying 12–17.9 hrs/week is a 94.59× multiplier for Grade C outcomes — providing a precise, data-backed threshold for academic support programs.
5	Sleep deprivation and academic stress form a self-reinforcing cycle: less sleep → higher stress → more late submissions → worse grades → greater pressure → less sleep.
6	Mathematical aptitude reliably predicts Science and Computer Science performance; English aptitude reliably predicts French and Social Studies — enabling data-backed educational pathway counseling.

9. Conclusion

The Data-Driven Education Monitor demonstrates the transformative potential of combining a robust data engineering pipeline with advanced business intelligence and conversational AI. By processing 1,356 student records through a three-tier Medallion Architecture on Microsoft Fabric, the project delivers a Star Schema data model powering nine specialized Power BI dashboards covering every dimension of student life — from academic scores and socio-economic background to digital engagement and psychological wellbeing.

The AI chatbot democratizes access to the data: stakeholders unfamiliar with BI tools can ask questions in plain English and receive precise, grounded answers through the Pandas agent or the RAG retrieval module. Together, these two layers — the BI dashboards and the chatbot — serve different stakeholder needs from the same underlying data platform.

Most critically, the project proves that student success cannot be optimized through any single intervention. Physical attendance, digital engagement, sleep quality, study habits, and socio-economic support are deeply interconnected variables that must be addressed as a unified system. The platform identifies at-risk students early, guides them toward appropriate educational pathways, and provides the data framework for evidence-based, unified intervention programs.

Final Positioning Statement

This project is an industry-style prototype demonstrating a complete, end-to-end data analytics platform — from raw CSV ingestion through PySpark transformations, Delta Lake storage, Star Schema modeling, nine Power BI dashboards, and an AI-powered conversational chatbot. It reflects the architecture, engineering patterns, and technologies used in enterprise Modern Data Stack environments, designed and delivered entirely by the student team. Core strengths demonstrated: Data Engineering: Microsoft Fabric + Apache Spark + Delta Lake + Medallion Architecture Business Intelligence: Power BI + DAX + Star Schema + AI Visuals (Key Influencers, Decomposition Tree) Applied AI: LangChain + RAG + Pandas Agent + Hybrid Routing + Flask API Business Thinking: Root-cause analysis + predictive modeling + actionable stakeholder insights